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Sound Event Detection using YOHO and Bottleneck Attention

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Abstract. Sound event detection (SED) plays a pivotal role in applications ranging from surveillance to healthcare, yet achieving high accuracy in dynamic acoustic environments remains challenging. This paper proposes an enhanced framework for SED by integrating the You Only Hear Once (YOHO) Convolutional Neural Network (CNN) architecture with a Bottleneck Attention Module (BAM) to improve feature localization and boundary detection in audio signals. Leveraging the Urban Sound Detection dataset, raw audio is preprocessed into mel spectrograms to capture spectral-temporal characteristics, which are fed into two models: (1) a baseline YOHO-based CNN and (2) a YOHO variant augmented with BAM. The attention mechanism dynamically weights channel-wise features, enabling the model to focus on salient audio regions while suppressing noise. Experimental results demonstrate that the attention-enhanced model achieves 90% test accuracy, outperforming the baseline YOHO model (87%) and exhibiting faster convergence. Key contributions include a systematic comparison of segmentation-by-classification approaches, a novel integration of BAM into YOHO for SED, and empirical validation of attention mechanisms in improving robustness against acoustic variability. This work underscores the potential of lightweight attention modules to enhance real-time sound event detection in resource-constrained IoT and edge computing applications, while future directions explore temporal modeling and multimodal fusion for further gains.

Keywords: Sound Event Detection, YOHO, Bottleneck Attention Module, Urban Sound Detection

1 Introduction

The rapid proliferation of audio-based technologies—from smart surveillance systems to healthcare monitoring tools—has underscored the critical need for robust sound event detection (SED) frameworks. SED aims to identify and temporally localize acoustic events within audio streams, enabling applications such as anomaly detection in industrial settings, real-time threat recognition in security systems, and context-aware automation in smart environments [1]. Despite

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its transformative potential, SED faces inherent challenges, including the dynamic nature of acoustic scenes, overlapping sound events, and the scarcity of labeled training data. Traditional approaches, such as Gaussian Mixture Models (GMMs) and Hidden Markov Models (HMMs), have proven inadequate for complex auditory environments due to their reliance on handcrafted features and limited ability to model temporal dependencies [2].

Recent advancements in deep learning have shifted the paradigm toward data-driven methods. Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) have demonstrated success in audio classification by leveraging spectral representations like mel spectrograms, which encode frequency and temporal information [3]. However, conventional CNNs often struggle with acoustic boundary detection—a key requirement for precise event localization—as they prioritize classification over temporal regression. Segmentation-by-classification approaches, which divide audio into fixed-duration frames for individual analysis, further introduce latency and computational overhead, hindering real-time deployment [4]. Additionally, the inherent variability of real-world audio, such as background noise and signal distortions, demands models capable of adaptive feature extraction.

To address these limitations, this work introduces a hybrid architecture combining the You Only Hear Once (YOHO) framework with a Bottleneck Attention Module (BAM). YOHO, inspired by YAMNet's efficiency in audio tagging, treats boundary detection as a regression task, enabling simultaneous event classification and temporal localization. The integration of BAM enhances this framework by dynamically recalibrating channel-wise feature responses, allowing the model to focus on salient spectral regions while suppressing irrelevant noise—a capability critical for environments with overlapping sounds. By processing mel spectrograms derived from the Urban Sound Detection dataset, our approach eliminates the need for manual feature engineering, streamlining the detection pipeline.

The contributions of this study are threefold:

- A novel YOHO-BAM framework is proposed that synergizes regression-based boundary prediction with attention-driven feature refinement.
- Comprehensive experiments demonstrating that BAM integration improves test accuracy by 3% (from 87% to 90%) and accelerates convergence compared to the baseline YOHO model.
- A scalable solution optimized for resource-constrained edge devices, validated on synthetic urban soundscapes with real-world noise interference.

This paper is structured as follows: Section II reviews existing SED methodologies. Section III details the dataset and mel spectrogram preprocessing pipeline. Sections IV, V, and VI elaborate on the core components of the proposed framework—the YAMNet architecture, YOHO framework, and Bottleneck Attention Module, respectively. Section VII presents the integrated YOHO-BAM system architecture. Section VIII discusses experimental results and comparative analysis. Finally, Section IX concludes the study and outlines future research directions, including temporal modeling and multimodal fusion.

2 LITERATURE REVIEW

The evolution of SED methodologies reflects a dynamic interplay between acoustic analysis and machine learning advancements. Contemporary SED systems aim to achieve two primary objectives: (1) identifying distinct acoustic classes within an audio stream and (2) precisely delineating their temporal boundaries. These capabilities are foundational for applications ranging from real-time surveillance to automated media indexing, where accurate event localization is as critical as classification. Historically, researchers have approached audio segmentation through two paradigms: distance-based segmentation and classification-driven segmentation.

Distance-based methods, such as Bayesian Information Criterion (BIC) and spectral divergence metrics, partition audio by detecting abrupt changes in acoustic features like energy or spectral coherence [5]. While effective for isolating homogeneous audio regions, these techniques often fail to classify events directly, necessitating secondary labeling steps. In contrast, segmentation-by-classification frameworks treat SED as a supervised learning problem, dividing audio into fixed-duration frames (e.g., 10–25 ms) and classifying each frame iteratively [6]. This approach aligns with modern deep learning practices but introduces computational bottlenecks due to sequential frame processing and post-hoc label smoothing.

Early SED systems borrowed heavily from speech and music recognition, employing Gaussian Mixture Models (GMMs) and Hidden Markov Models (HMMs) to model temporal dependencies [7]. However, these methods exhibit limitations in polyphonic environments, where overlapping sound events lack the structured temporal patterns of phonemes or musical notes. The advent of deep learning shifted focus toward neural architectures capable of joint feature extraction and classification. Bidirectional Long Short-Term Memory (BiLSTM) networks, for instance, demonstrated promise in capturing contextual dependencies across time-series audio data [8]. Yet, their sequential processing nature limits scalability for real-time applications.

Recent innovations highlight the efficacy of hybrid architectures. Temporal Convolutional Networks (TCNs), leveraging dilated convolutions, outperform BiLSTMs in latency-sensitive tasks by processing audio frames in parallel [9]. Concurrently, Convolutional Recurrent Neural Networks (CRNNs) merge the spatial feature extraction prowess of CNNs with the temporal modeling of RNNs, achieving state-of-the-art results on benchmark datasets like DCASE [10]. Despite these advances, CRNNs remain hindered by post-processing overheads, as frame-level predictions require aggregation into coherent event boundaries.

A pivotal challenge in SED lies in transitioning from frame-wise classification to end-to-end boundary prediction. Current trends emphasize attention mechanisms and transformer architectures to bypass post-processing. For example, self-attention layers enable models to weigh salient spectral regions dynamically, enhancing both detection accuracy and temporal resolution [11]. Such advancements align with the broader shift toward lightweight, interpretable models op-

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timized for edge deployment—a necessity for applications like bioacoustic monitoring and industrial IoT [12].

Despite progress, critical gaps persist. Existing frameworks often prioritize classification over precise temporal localization, leading to boundary ambiguity in dynamic environments. Computational overhead from post-processing steps further limits real-time deployment, while performance degradation under acoustic variability (e.g., background noise) remains unresolved. This work addresses these challenges by integrating YOHO, a regression-based CNN, with Bottleneck Attention, enabling simultaneous event classification and boundary prediction without post-processing. By bridging these gaps, our approach enhances robustness and efficiency, aligning with the demands of modern IoT ecosystems.

3 METHODOLOGY

3.1 Dataset Collection

To validate the proposed framework, this study employs the UrbanSound8K dataset, a benchmark resource widely adopted in SED research for its realistic simulation of urban acoustic environments. Synthesized using the Scaper toolkit, the dataset replicates complex auditory scenes by strategically overlaying isolated sound events—such as Car Horn, Drilling, and Gun Shot—onto diverse background noise profiles, including street ambience and indoor settings. This synthetic design addresses the scarcity of labeled real-world audio data while enabling precise control over acoustic variables like signal-to-noise ratio (SNR) and spatial effects, critical for training models robust to real-world variability.

As shown in Fig. 1, the dataset comprises 10 distinct urban sound classes, spanning both stationary (e.g., Air Conditioner, Engine Idling) and transient (e.g., Dog Bark, Jackhammer) events, ensuring comprehensive coverage of acoustic phenomena relevant to IoT and smart city applications. Each audio clip is standardized to a 16 kHz sampling rate and mono-channel configuration to maintain input consistency, while temporal annotations for event boundaries facilitate supervised learning of joint classification and localization tasks. Organized into predefined training, validation, and testing subsets totaling 30 hours of audio, the dataset mitigates overfitting risks and ensures reproducibility—a key requirement for benchmarking against state-of-the-art SED frameworks. By mirroring challenges such as overlapping sounds and background interference, this dataset aligns with the gaps identified in existing literature, providing a rigorous foundation for evaluating the proposed YOHO-BAM architecture’s ability to enhance temporal precision and noise robustness.

3.2 Audio Processing

In our research, audio processing plays a crucial role as we aim to extract mel spectrograms from raw audio data for input into both the attention-based YAM-net model and the YOHO model. To derive the mel spectrogram from raw audio,

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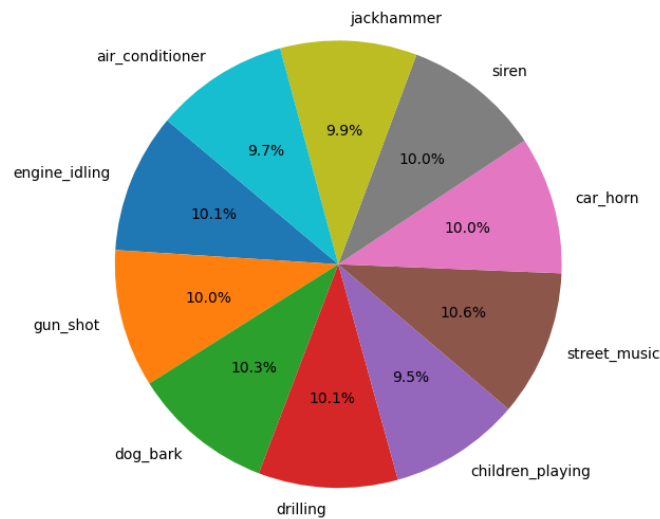


Fig. 1. Classwise Dataset distribution

several sequential steps are employed. Initially, the raw audio is segmented into short frames with overlapping sections. Subsequently, windowing functions such as the Hamming or Hann window are applied to each frame to address spectral leakage. The Short-Time Fourier Transform (STFT) is then computed for each frame, resulting in its frequency domain representation. Following this, the Mel Filterbank is directly applied to the magnitude spectrum obtained from the STFT, without employing normalization or logarithmic compression. This process yields a series of mel-filtered spectrograms, with each frame capturing the spectral characteristics of the audio signal across time. These mel spectrograms serve as input features for our model.

Figure 3 depicts the time domain representation of an audio sample, showcasing the variation in air pressure over a specified period. In contrast, Figure 4 illustrates the mel-spectrogram representation of the same audio sample. Here, we gain insights into the different frequencies associated with the sound sample and how they change over time. This mel spectrogram serves as the input to the model for our research study.

4 YAMNET ARCHITECTURE

YAMNet, short for 'Yet Another Multitask Network', is a deep learning model developed by Google's Machine Perception team specifically for sound analysis tasks. In our research work, the model architecture we are using is the YAMNet model. YAMNet [14] uses the MobileNetV1 architecture as its foundation. Therefore, in our study, we train YAMNet on evenly distributed data and achieve performance gains without tables. Before the feature extraction stage, we resam-

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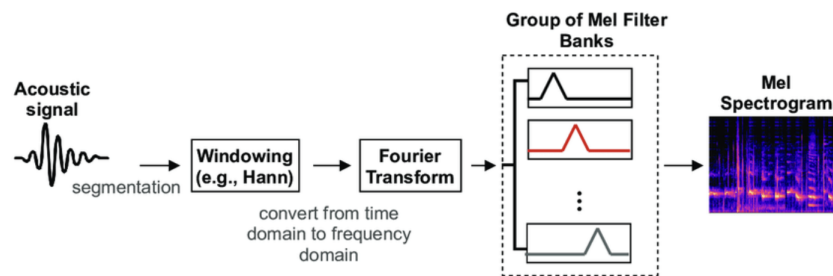


Fig. 2. Steps to get mel-spectrogram [15]

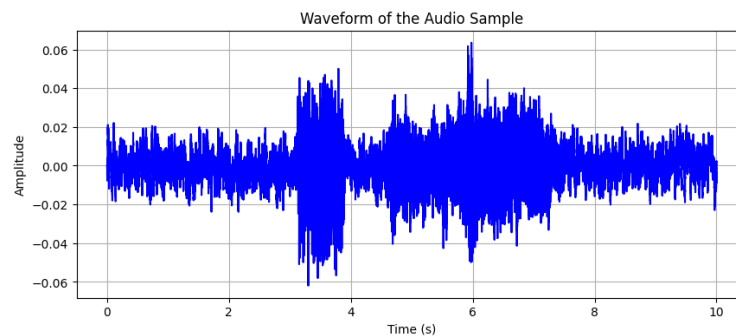


Fig. 3. Raw Audio Signal

ple the audio input to 48,000 Hz with a single- channel configuration. Furthermore, being a deep learning- based model, YAMNet automatically extracts audio features thanks to its feature extraction layer. In our model we have used bottleneck attention block after each CNN layer to effectively obtain an hierarchial attention.

5 YOHO

YOHO operates as a convolutional neural network (CNN), with its architecture split into two parts: the upper section containing the original layers from the YAMNet architecture, and the lower section housing additional layers. Input features are log-mel spectrograms, with dimensions determined by the audio example's duration and mel spectrogram specifications. For urban sound event detection, input comprises 801 time steps and 64 frequency bins. Reshaping the mel spectrogram to $801 \times 64 \times 1$, a 2D convolution with a stride of 2 is performed, halving both time and frequency dimensions. The MobileNet architecture employs depthwise-separable convolutions with 3×3 filters, followed by pointwise convolutions with 1×1 filters. Except for the final layer, convolutions

Sound Event Detection using YOHO and Bottleneck Attention

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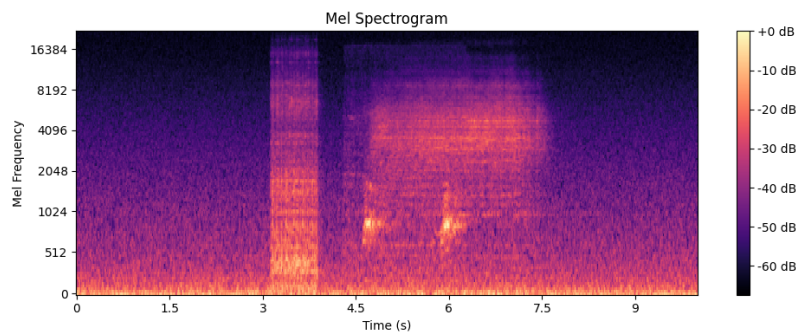


Fig. 4. Mel Spectrogram For the Audio Signal in Fig 3

are activated with ReLu and batch normalization is applied. Each stride of 2 leads to a reduction in time and frequency dimensions.

6 BOTTLENECK ATTENTION

To improve the performance of the YAMNet model that we used with the YOHO algorithm, we used attention block (bottleneck attention). In bottleneck attention the inference is done along spatial and the channel. It is called bottleneck as it is placed at the bottleneck of each model while downsampling the model during the convolution operation. The training process involves optimizing various channel weights through a cost function, leading to the automatic derivation of feature channel coefficients. The integration of an attention module aids the model in capturing intermediate features more efficiently [13].

7 SYSTEM ARCHITECTURE

The Methodology of this work starts from the data collection step. We have audio samples as the raw data we need to get the data in a different format to feed it to our model. For our reserach work we are using the mel spectrogram. To get that we first follow the audio processing steps. First we convert the audio into the overlapping frames and then the windowing function is applied. Then short term fourier transform is done to get the spectrogram. We then apply the mel filter bank to get the mel spectrogram. We then use this spectrogram as the input feature for the two models that we are using in our research work. The first model is the Yoho model which is a CNN based model which uses YAMNet architecture, the second model is the Yoho model with the bottleneck attention. In Figure 5, the above flow is explained.

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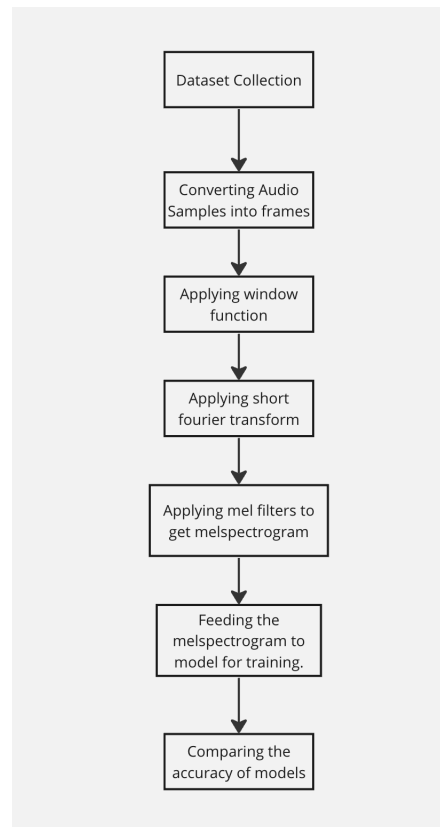


Fig. 5. System Architecture

8 EXPERIMENTS AND RESULTS

8.1 Analysis using Yoho Model

Table 1 displays the accuracy results of the Yoho-based CNN model on both the training and test datasets. Additionally, Figure 6 & 7 illustrates the progressive increase in model accuracy across epochs. Our experiments involved running the model for approximately 24 epochs, during which we observed that the accuracy plateaued and the model reached convergence. Furthermore, there was negligible change in the training loss and accuracy beyond this point, indicating saturation in model performance.

8.2 Analysis using Yoho with attention Model

Table 2 shows the accuracy of the Attention-based model on both the training and test datasets. Figure 8 illustrates the progressive increase in the accuracy

Table 1. Results on YOHO Based Model.

Data	Accuracy (%)
Train data	95
Validation data	90
Test data	87

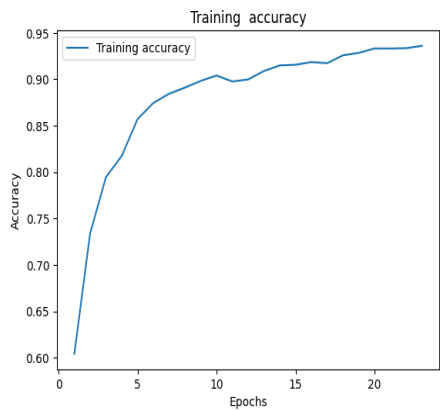


Fig. 6. Accuracy vs Epoch for train data

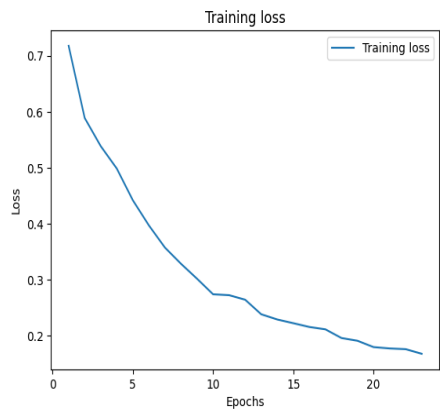


Fig. 7. Loss vs Epoch for train data

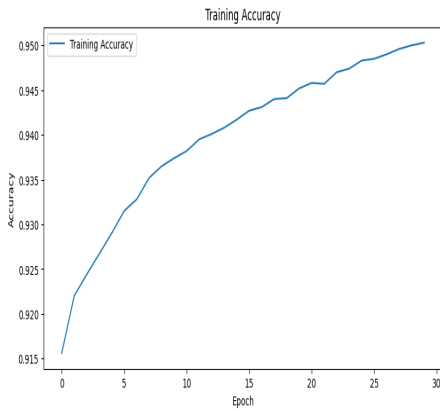


Fig. 8. Accuracy vs Epoch for train data

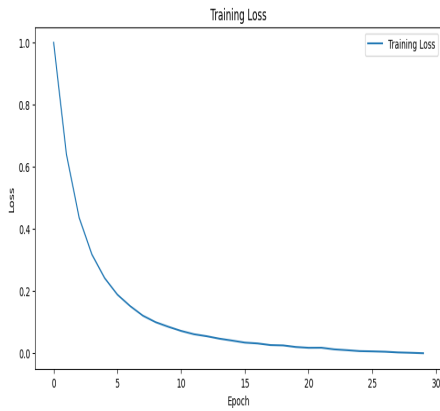


Fig. 9. Loss vs Epoch for train data

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Table 2. Results on Attention-Based Model.

Data	Accuracy (%)
Train data	0.93
Validation data	0.88
Test data	0.90

of the model over each epoch. Our experiment involved training the model for approximately 24 epochs, revealing that the accuracy of the model plateaued, indicating convergence. Furthermore, there was minimal change in training loss and accuracy beyond this point, indicating stability in model performance. In particular, the Bottleneck Attention-based model exhibits slightly superior performance compared to the YOHO-based model. Figure 9 further demonstrate the faster convergence rate of the attention-based model.

9 CONCLUSION AND FUTURE WORK

This study investigated the integration of bottleneck attention with the YOHO architecture for SED on the TUT-Urban dataset. The attention-enhanced model outperformed the baseline YOHO, achieving 90% test accuracy (vs. 87% for YOHO) and faster convergence, validating the efficacy of attention mechanisms in refining feature focus and robustness. While the results highlight the promise of attention-driven CNNs for SED, limitations such as reliance on synthetic data and computational inefficiencies in frame-based processing warrant further exploration. Future work could extend this research by incorporating temporal modeling (e.g., transformers or RNNs) to capture long-range dependencies, exploring multimodal fusion with text or visual data, and investigating advanced attention variants like multi-head mechanisms. Additionally, optimizing the model for edge deployment and leveraging semi-supervised learning to reduce annotation dependency could enhance real-world applicability. These advancements would strengthen SED systems for use in healthcare, environmental monitoring, and smart infrastructure.

References

1. Parekh, S., Essid, S., Ozerov, A., Duong, N.Q.K., Pérez, P., Richard, G.: Weakly supervised representation learning for audio-visual scene analysis. *IEEE/ACM Trans. Audio Speech Lang. Process.* **28**, 416–428 (2019).
2. Theodorou, T., Mporas, I., Fakotakis, N.: An overview of automatic audio segmentation. *Int. J. Inf. Technol. Comput. Sci.* **6**(11), 1 (2014).
3. Venkatesh, S., Moffat, D., Miranda, E.R.: You only hear once: a YOLO-like algorithm for audio segmentation and sound event detection. *Appl. Sci.* **12**(7), 3293 (2022).

Sound Event Detection using YOHO and Bottleneck Attention 11

4. Mesaros, A., Heittola, T., Virtanen, T.: TUT database for acoustic scene classification and sound event detection. In: Proc. 24th Eur. Signal Process. Conf. (EU-SIPCO), pp. 1128–1132 (2016).
5. Salamon, J., Bello, J.P., Farnsworth, A., Robbins, M., Keen, S., Klinck, H., Kelling, S.: Towards the automatic classification of avian flight calls for bioacoustic monitoring. PLoS One **11**(11), e0166866 (2016).
6. Mesaros, A., Heittola, T., Virtanen, T.: TUT database for acoustic scene classification and sound event detection. In: Proc. 24th Eur. Signal Process. Conf. (EU-SIPCO), pp. 1128–1132 (2016).
7. Gimeno, P., Viñals, I., Ortega, A., Miguel, A., Lleida, E.: Multiclass audio segmentation based on recurrent neural networks for broadcast domain data. EURASIP J. Audio Speech Music Process. **2020**, 1–19 (2020).
8. Lemaire, Q., Holzapfel, A.: Temporal convolutional networks for speech and music detection in radio broadcast. In: Proc. 20th Int. Soc. Music Inf. Retr. Conf. (ISMIR), pp. –, 4–8 Nov 2019. Int. Soc. Music Inf. Retr. (2019).
9. Venkatesh, S., Moffat, D., Kirke, A., Shakeri, G., Brewster, S., Fachner, J., Odell-Miller, H., Street, A., Farina, N., Banerjee, S., et al.: Artificially synthesising data for audio classification and segmentation to improve speech and music detection in radio broadcast. In: Proc. IEEE Int. Conf. Acoust. Speech Signal Process. (ICASSP), Toronto, Canada, pp. 636–640 (2021).
10. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, L., Polosukhin, I.: Attention is all you need. Adv. Neural Inf. Process. Syst. **30** (2017).
11. Yang, Z., Cen, J., Liu, X., Xiong, J., Chen, J., Honghua: Research on bearing fault diagnosis method based on transformer neural network. Meas. Sci. Technol. **33**, 10.1088/1361-6501/ac66c4 (2022).
12. Park, J., Woo, S., Lee, J.-Y., Kweon, I.S.: Bam: Bottleneck attention module. arXiv preprint arXiv:1807.06514 (2018).
13. Mahum, R., Irtaza, A., Javed, A., Mahmoud, H.A., Hassan, H.: DeepDet: YAMNet with BottleNeck Attention Module (BAM) for TTS synthesis detection. EURASIP J. Audio Speech Music Process. **2024**(1), 18 (2024).
14. Kuzminykh, I., Shevchuk, D., Shiaeles, S., Ghita, B.: Audio interval retrieval using convolutional neural networks. In: Proc. Int. Conf. Next Gen. Wired/Wireless Netw., pp. 229–240 (2020).
15. Pham, P., Li, J., Szurley, J., Das, S.: Eventness: Object detection on spectrograms for temporal localization of audio events. In: Proc. IEEE Int. Conf. Acoust. Speech Signal Process. (ICASSP), pp. 2491–2495 (2018).